Bin Zhao

Ph.D. Student in Materials Science and Engineering | Solid State Ionics Lab | Westlake University

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RESEARCH INTERESTS

Electrochemical ion intercalation, defect chemistry, and ion transport in functional oxide thin films in room-temperature aqueous environments. Oxide thin-film preparation by pulsed laser deposition (PLD) and magnetron sputtering, combined with operando electrochemical and spectroscopic characterization.

EDUCATION

Westlake University, Hangzhou, China | Sep 2024 - Present

Ph.D. student, Materials Science and Engineering, School of Engineering; joint Ph.D. program with Zhejiang University. Advisor: Prof. Qiyang Lu.

Zhengzhou University, Zhengzhou, China | Sep 2020 - Jun 2024

B.E., Metallic Materials and Forming Processing, School of Materials Science and Engineering. GPA: 3.70/4.00; Rank: 8/125.

PUBLICATIONS

1. Bin Zhao, Luhan Wei, Xi Wang, Zihan Xu, Ying Lu, Jieping Zheng, Lingzhi Huang, and Qiyang Lu. Aerobic Dehydrogenation by Coupled Electrochemical Reactions Governs Room-Temperature Proton Defect Chemistry in Functional Oxides. Journal of the American Chemical Society, 148(40), 43401-43410 (2026). DOI
2. Bin Zhao, Luhan Wei, Jieping Zheng, Ying Lu, Haowen Chen, Kaichuang Yang, and Qiyang Lu. Quantifying Chemical Capacitance and Diffusion Coefficients in Hydrogenated La0.6Sr0.4FeO3−δ via Operando Spectro-Electrochemical Characterizations. Nano Letters, 26(4), 1578-1586 (2026). DOI
3. Jieping Zheng, Kaichuang Yang, Ying Lu, Ziyun Zhang, Rui Wang, Luhan Wei, Bin Zhao, Hui Zhang, and Qiyang Lu. Elucidating Ionic Programming Thermodynamics of Oxide-Ion-Based Electrochemical Random-Access Memory Devices. Small, e75805 (2026). DOI
4. Luhan Wei, Haowen Chen, Zihan Xu, Yuxin Hu, Bin Zhao, Ying Lu, Nan Zhang, and Qiyang Lu. Quantifying Hydrogen Chemical Diffusivity in NdNiO3 Thin Films through Operando Multimodal Measurements. Nano Letters, 25(15), 6348-6355 (2025). DOI

PATENT

Xiaohe Miao, Bin Zhao, Qiyang Lu. Oxygen-Free Circulating Liquid Cell and In-Situ Electrochemical Characterization Method for High-Resolution X-Ray Diffraction on Horizontal Goniometer. Chinese Patent No. ZL 2025 1 1639587.5 (Issued 2026).

HONORS & AWARDS

- SSI-25 Student Travel Support Grant, 25th International Conference on Solid State Ionics (Jul 2026).
- Orient Gene Scholarship, Westlake University (Oct 2026).
- Outstanding Graduate Student, Zhejiang University (Oct 2026).
- Merit Student of Henan Province (Apr 2023).

CONFERENCE PRESENTATIONS

Oral Presentation, 25th International Conference on Solid State Ionics (SSI-25), Singapore, July 2026.

EXPERIMENTAL & COMPUTATIONAL SKILLS

Fabrication: PLD, magnetron sputtering, lithography, spin coating, hydrothermal and solvothermal synthesis.

Electrochemistry: CV, CA, EIS, and operando electrochemical measurement.

Characterization: Operando UV-vis, APXPS, Raman, high-resolution XRD; AFM, TOF-SIMS, synchrotron XAS at SSRF.
Programming: Python.